Tasks 1: COCA Dataset Preprocessing

Dataset Overview		
Metric	**Value**	**Percentage**
Total Scans	787	100%
Zero Calcium	340	43%
Non-zero Calcium	447	57%

Calcium Voxels Distribution	
Statistic	**Value**
Mean	587.5
Standard Deviation	1351.6
Median	33
25th Percentile	0
75th Percentile	464.5
Minimum	0
Maximum	11,678
Zero-Calcium %	43.20%

Preprocessing & Augmentation

The preprocessing pipeline for the COCA dataset was designed to produce spatially and intensity-normalized cardiac CT volumes suitable for downstream coronary atlas registration. All 787 non-contrast CT scans were resampled to a uniform 0.7 × 0.7 × 3.0 mm voxel spacing following the PrediCT project recommendation, balancing in-plane resolution with the anisotropic nature of cardiac CT acquisition where inter-slice spacing is inherently coarser than in-plane resolution. HU windowing was applied with a center of 400 HU and width of 1000 HU (range: −100 to 900 HU), a range chosen to maximally contrast calcium deposits (200–900 HU) against surrounding soft tissue. Data augmentation was restricted to geometrically conservative transforms, left/right flips (p=0.50), small rotations up to ±15° (p=0.50), zoom scaling between 0.85–1.15× (p=0.30), Gaussian noise (σ=0.05, p=0.20), and intensity scaling (±10%, p=0.20). All geometric transforms were applied identically to both CT volumes and segmentation masks using MONAI dictionary transforms, and a final intensity clamp ensured all augmented values remained within the normalized [0, 1] range.

Split Strategy & Project 3 Preparation

The dataset was partitioned into training (70%), validation (15%), and test (15%) sets based on five CAC categories:

CAC Category Distribution		
Category	**Voxel Range**	**%**
Zero	=0	43.2%
Minimal	1-100	14.6%
Mild	101-500	18.2%
Moderate	501-1500	11.8%
Severe	>1500	12.2%

While the dataset exhibits notable class imbalance with 43.2% zero-calcium scans, weighted resampling was deliberately omitted as this pipeline targets Project 3 (Physics-Based Simulation — Coronary Atlas Registration). All resampled volumes and binary calcium masks are stored as compressed NIfTI files and indexed in split_index.csv, ready for direct ingestion by the SimpleITK-based registration pipeline.

DataLoader

The PyTorch DataLoader was designed as a memory-efficient, augmentation-ready pipeline for training calcium scoring models. Preprocessed scans are stored on disk as compressed NIfTI files and loaded lazily per batch, avoiding the memory overhead of loading the full 787-scan dataset into RAM. A custom CachedCOCADataset class reads scan-level metadata from the split index CSV, loads the corresponding image and mask pair on demand, and applies a configurable augmentation pipeline during training. Augmentations include random horizontal and vertical flipping, rotation up to 15 degrees, random intensity scaling within ±10%, and Gaussian noise injection, all implemented as composable transforms applied consistently to both image and mask. A class-weighted sampler addresses the dataset imbalance between Zero (43%) and non-zero CAC categories by upsampling under-represented severity classes during training.

Task 2: Project 3 (Physics-Based Simulation) - Coronary Atlas Registration

Background

The goal of Part 2 was to register a publicly available coronary anatomy atlas from the ImageCAS CCTA dataset to 30 non-contrast CT scans from the COCA test split, producing approximate vessel territory masks in patient space for downstream calcium placement modelling. The registration task presents a fundamental cross-modal challenge: the ImageCAS atlas is a contrast-enhanced CCTA scan in which coronary artery lumens appear bright due to iodine contrast agent, while the COCA target scans are non-contrast acquisitions in which the same vessels are invisible, appearing isodense with myocardium at approximately 50–80 HU. Direct intensity-based metrics such as Mean Squared Error or Normalised Cross-Correlation are therefore inapplicable, as they assume either identical or linearly related intensity distributions across the two images.

Registration Algorithm

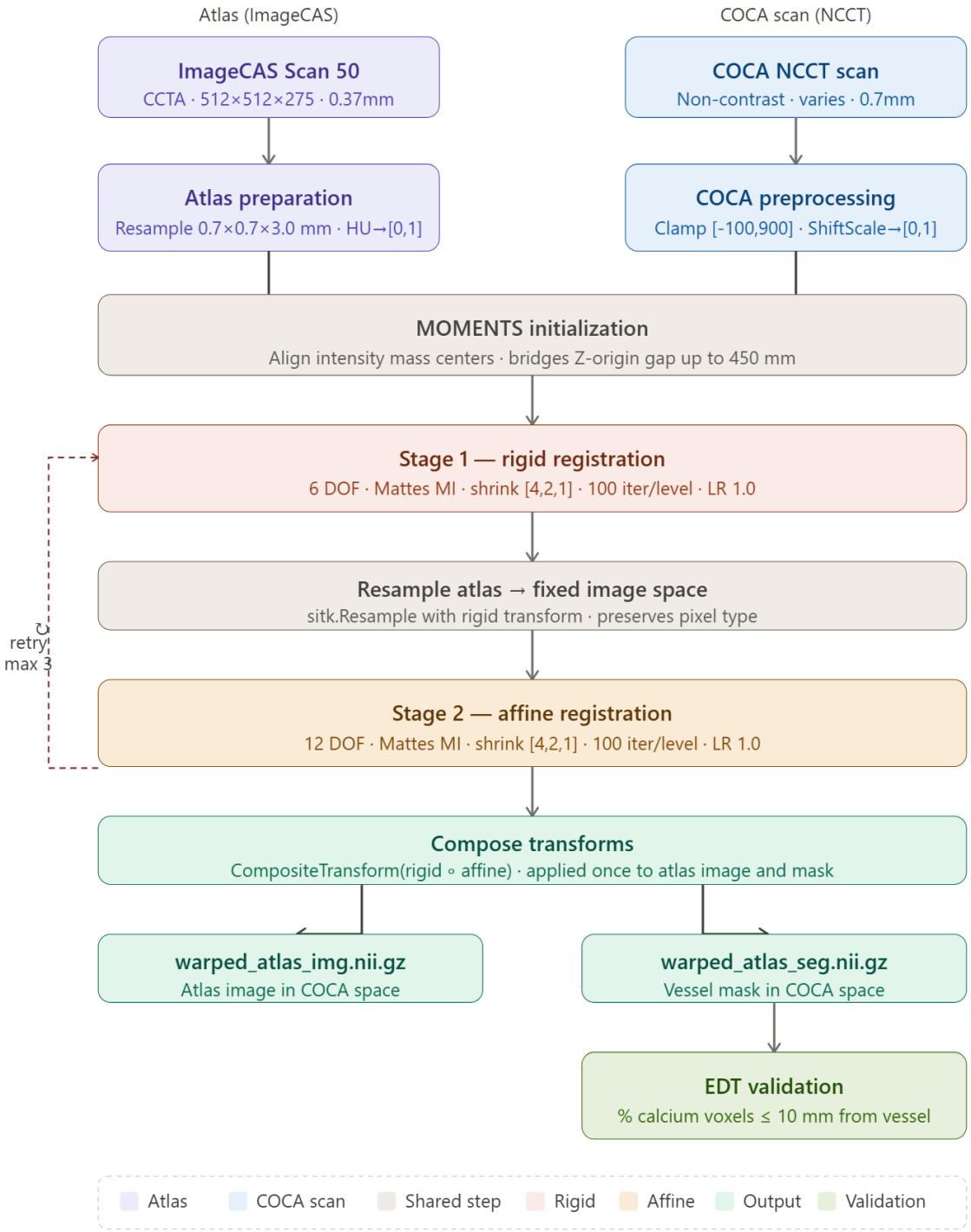
1. Mutual Information (MI) was selected as the similarity metric. MI measures statistical dependency between the full intensity distributions of the two images rather than comparing individual voxel values.

2. Registration was implemented in SimpleITK using a two-stage rigid-then-affine pipeline. The rigid stage (6 degrees of freedom) corrects for global differences in patient positioning and scanner table orientation across acquisitions, while the affine stage (12 degrees of freedom) additionally accounts for anisotropic scaling and shear arising from differences in heart size, shape, and breath-hold position between the atlas donor and each target patient.

3. A critical implementation challenge was the mismatch in physical coordinate origins between the ImageCAS atlas and the COCA scans. The atlas carried a Z-origin of approximately +44 mm, while COCA scans had Z-origins ranging from −200 to −400 mm, a gap of up to 450 mm in physical space. Initialising registration from geometric bounding-box centers (GEOMETRY mode) placed the atlas entirely outside the COCA volume before optimisation began, causing the MI metric to receive zero valid samples and immediately fail. This was resolved by switching to MOMENTS initialisation, which aligns the intensity-weighted centers of mass of the two images. Since the heart is the dominant high-intensity structure in both modalities, its intensity mass center is consistent across acquisitions regardless of scanner-assigned coordinate origins, allowing MOMENTS to correctly bridge the coordinate gap without any explicit anatomical landmark detection.

4. A three-level multi-resolution pyramid with shrink factors [4, 2, 1] and Gaussian smoothing sigmas [2.0, 1.0, 0.0] mm was applied at both stages to avoid local minima and reduce computation at coarser resolution levels. The GradientDescentLineSearch optimizer ran for 100 iterations per level with a learning rate of 1.0 and early stopping when the convergence criterion fell below 1×10^{-6} over a 10-iteration window.

5. Validation quantified the anatomical proximity of registered vessel territories to actual calcium deposits using a Euclidean Distance Transform (EDT) approach. For each successfully registered scan, the warped vessel mask was binarised and the EDT was computed using scipy.ndimage.distance_transform_edt with physical spacing correction, assigning every voxel in the volume its distance in millimetres to the nearest vessel voxel. The calcium segmentation mask was then used to index into this distance map, and the percentage of calcium voxels with a distance of 10 mm or less was reported as the primary validation metric.



Experiment 1 - Baseline

To establish a baseline, ImageCAS Scan 100 was selected as the initial atlas candidate on the basis of vessel voxel count; it exhibited the highest number of labelled vessel voxels (137,547) among the candidates inspected. This heuristic selection formed Experiment 1, against which the systematic four-candidate comparison of Experiment 2 was evaluated.

Experiment 2 - Atlas Selection

Three atlas candidates (ImageCAS Scans 1, 10, 50) were systematically evaluated against the randomly selected Scan 100 baseline on a representative 8-scan subset (2 per CAC category). Each candidate was evaluated on a representative 8-scan subset, two scans per CAC severity category, using three selection criteria:

- Registration success rate
- Mean MI
- Mean calcium-to-vessel distance

Scan 50 was selected as the final atlas on the basis of achieving 8/8 successful registrations (versus 5/8 for Scan 1, which produced overlap failures on three scans despite nominally higher validation scores on the cases it completed), the highest mean MI metric across the subset (0.245, indicating the greatest statistical compatibility with the COCA population intensity distribution), and the lowest mean calcium-to-vessel distance (8.7 mm).

Atlas	Vessel Voxels	Mean % ≤10mm	Passing (>70%)	Mean Time (s)	Mean |MI|
Scan 1	5,285	70.6%	3/5	17.9s	0.213
Scan 10	4,256	42.4%	2/6	17.5s	0.241
Scan 50	**4,368**	**66.5%**	**5/8 ✓**	**21.0s**	**0.245**
Scan 100	4,772	55.2%	3/7	16.6s	0.173

Final Results: Experiment 3

Experiment 3 ran the full pipeline using Scan 50 as the atlas. All 30 registrations completed successfully (100% success rate). The median validation score of 71.6% calcium within ±10 mm met the >70% target; the mean of 68.5% fell 1.5% short, representing an improvement of +18.1 percentage points over the Experiment 1 baseline (Scan 100, mean 50.4%).

Metric	Minimal	Mild	Moderate	Severe
Scans	7	7	7	9
Passing > 70%	6/7 (86%)	5/7 (71%)	2/7 (29%)	4/9 (44%)

Mean % ≤ 10 mm	85.7%	78.8%	54.6%	58.0%
Mean dist to vessel	5.6 mm	5.9 mm	10.7 mm	10.2 mm

Overall	Scan 100 (n=25)	Scan 50 (n=30)
Registration success	24/25 (96%)	30/30 (100%)
Passing > 70%	8/25 (32%)	17/30 (57%)
Mean % ≤ 10 mm	50.4%	68.5%
Median % ≤ 10 mm	44.0%	71.6% (meets target)
Mean dist to vessel	12.9 mm	8.3 mm
Mean time per scan	15.7 s	22.1 s
Total time (30 scans)	—	11.1 minutes

Conclusion

A systematic performance gap was observed between low- and high-burden CAC categories. Minimal and Mild cases achieved mean validation scores of 85.7% and 78.8% respectively, while Moderate and Severe cases reached only 54.6% and 58.0%. In early-stage coronary artery disease, calcium deposits are small, localised plaques that sit on vessel walls close to the position predicted by the healthy atlas. In advanced disease, years of atherosclerotic remodelling displace the vessel wall itself, so the healthy atlas vessel position diverges from the diseased vessel geometry regardless of registration accuracy. This finding directly motivates the hemodynamic probability map approach proposed in the main PrediCT project: a spatial prior that encodes where calcium is biologically likely to deposit, accounting for disease progression, is more robust for synthetic data generation than a hard vessel boundary derived from a single healthy atlas registered to a potentially heavily diseased patient.